

Mount Sinai Launches AI-Powered Clinical Trial-Matching Platform to Expand Access to Cancer Research

- Thousands of individuals with the condition are likely undiagnosed.
- Study identifies underutilization of genetic testing as a major contributor to missed diagnoses.
- Findings underscore the growing importance of genetic testing as precision therapies advance toward patients.

New York, NY (January 08, 2026) — The Mount Sinai Tisch Cancer Center (TCC) has launched a new artificial intelligence (AI) platform to help connect more cancer patients across the Mount Sinai Health System to life-saving clinical trials. The platform, which was created by the AI company Triomics, is called PRISM, and is powered by Triomics’ OncoLLM, a large language model-based pipeline built especially for cancer care. As an early adopter of this new technology, Mount Sinai becomes the first National Cancer Institute (NCI)-designated Comprehensive Cancer Center in New York City to deploy this oncology-specific AI tool for systemwide clinical trial matching.

The deployment marks a major step forward in expanding access to innovative cancer research, accelerating clinical trial enrollment, and ensuring patients across the Health System are considered for cutting-edge treatment options earlier in their care.

“Clinical trials are essential to advancing cancer care, but too often patients and their treating physicians are not aware of studies that may be appropriate for them,” said Karyn Goodman, MD, MS, Professor and Vice Chair of Clinical Research in the Department of Radiation Oncology at the Icahn School of Medicine at Mount Sinai and Associate Director of Clinical Research at TCC. “By deploying an AI platform trained specifically for oncology, we can identify trial opportunities earlier, more consistently, and more equitably, allowing clinicians to focus on meaningful conversations with patients rather than manual chart review.”

Clinical trial matching is the process of determining which research studies a patient may be eligible for based on their electronic health records, diagnosis, medical history, and specific clinical characteristics. Traditionally, this work has relied on manual review of lengthy trial protocols and fragmented medical records, a time-intensive process that can sometimes delay or limit patient access to promising new therapies.

Historically, trial participation has been concentrated at flagship academic sites. With systemwide deployment, patients seen at Mount Sinai Queens, Mount Sinai Brooklyn, Mount Sinai South Nassau, Mount Sinai Morningside, Mount Sinai West, and other sites have the same accessibility to available trials as those treated at The Mount Sinai Hospital.

This initiative directly advances Mount Sinai’s mission as an NCI-designated Comprehensive Cancer Center, strengthening systemwide access to clinical trials, accelerating research integration across care settings, and supporting improved outcomes for the diverse communities it serves. By expanding the pool of patients who can be considered for research at the right moment in their care, Mount Sinai is positioned to meet, and potentially exceed, national benchmarks for trial participation.

The successful deployment reflects close collaboration among Triomics, TCC, and Mount Sinai Research IT, combining technical innovation, clinical leadership, and robust governance to bring clinical trials closer to the bedside. Mount Sinai plans to evaluate outcomes from this deployment and share findings through future peer-reviewed publications and national scientific meetings.

For more Mount Sinai artificial intelligence news, visit: <https://icahn.mssm.edu/about/artificial-intelligence>.

About the Icahn School of Medicine at Mount Sinai

The Icahn School of Medicine at Mount Sinai is internationally renowned for its outstanding research, educational, and clinical care programs. It is the sole academic partner for the seven member hospitals* of the Mount Sinai Health System, one of the largest academic health systems in the United States, providing care to New York City’s large and diverse patient population.

The Icahn School of Medicine at Mount Sinai offers highly competitive MD, PhD, MD-PhD, and master’s degree programs, with enrollment of more than 1,200 students. It has the largest graduate medical education program in the country, with more than 2,600 clinical residents and fellows training throughout the Health System. Its Graduate School of Biomedical Sciences offers 13 degree-granting programs, conducts innovative basic and translational research, and trains more than 560 postdoctoral research fellows.

Ranked 11th nationwide in National Institutes of Health (NIH) funding, the Icahn School of Medicine at Mount Sinai is among the 99th percentile in research dollars per investigator according to the Association of American Medical Colleges. More than 4,500 scientists, educators, and clinicians work within and across dozens of academic departments and multidisciplinary institutes with an emphasis on translational research and therapeutics. Through Mount Sinai Innovation Partners (MSIP), the Health System facilitates the real-world application and commercialization of medical breakthroughs made at Mount Sinai.

** Mount Sinai Health System member hospitals: The Mount Sinai Hospital; Mount Sinai Brooklyn; Mount Sinai Morningside; Mount Sinai Queens; Mount Sinai South Nassau; Mount Sinai West; and New York Eye and Ear Infirmary of Mount Sinai.*

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About the Mount Sinai Health System

Mount Sinai Health System is one of the nation’s leading integrated academic health systems and one of the largest in the New York metropolitan area. Its comprehensive system includes seven hospitals, more than 400 outpatient practices, over 600 research and clinical laboratories, the Icahn School of Medicine at Mount Sinai, the Graduate School of Biomedical Sciences, and the Mount Sinai Phillips School of Nursing. Together, the Health System comprises approximately 48,000 employees, more than 9,000 physicians, and 8,600 nurses.

As a leading learning health system, Mount Sinai combines clinical expertise with scientific discovery to improve patient care while training the next generation of health care and biomedical leaders. The Health System provides care across every stage of life, from prenatal care through geriatrics, while advancing personalized medicine through artificial intelligence, data science, and biomedical research.

Mount Sinai is consistently recognized among the nation’s leading academic health systems for patient care, research, and education. The Mount Sinai Hospital is ranked No. 1 in New York by Newsweek and No. 5 on the magazine’s World’s Best Hospitals list. The Icahn School of Medicine at Mount Sinai ranks No. 11 among U.S. medical schools and No. 1 among freestanding medical schools for National Institutes of Health funding, reflecting the strength of its scientific enterprise and leadership in biomedical research.